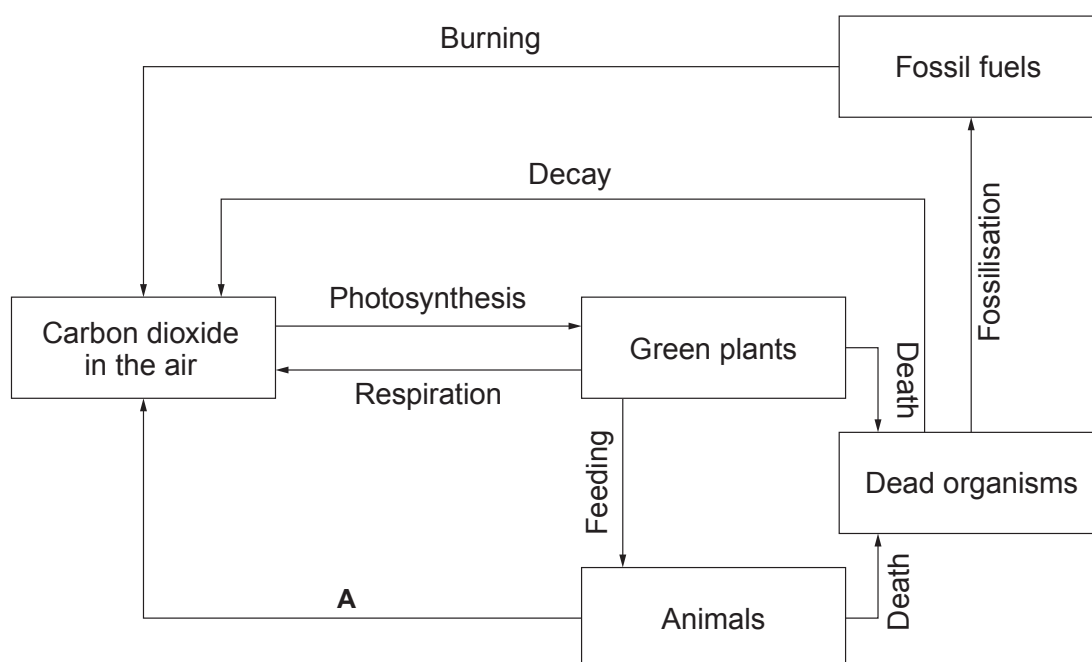


2. The diagram shows the carbon cycle.



(a) Use the diagram and your knowledge to answer the following questions.

(i) Circle the name of the process shown by label **A**.

[1]

feeding                  photosynthesis                  respiration

(ii) Tick (✓) the boxes next to the **three** correct statements.

[3]

Photosynthesis uses carbon dioxide from the atmosphere.

☐

Decomposing dead plants adds carbon dioxide to the atmosphere.

☐

Fossilising dead plants adds carbon dioxide to the atmosphere.

☐

Respiration adds carbon dioxide to the atmosphere.

☐

Burning dead plants removes carbon dioxide from the atmosphere.

☐

When green plants feed they add carbon dioxide to the atmosphere.

☐

(iii) State **two** ways that human activity has caused carbon dioxide levels to increase.

[2]

1. ....

2. ....



- (b) The table shown below is taken from a climate change website.

It gives the mass of carbon dioxide produced by a person travelling to work either by car or bus.

|                                                                   |      |
|-------------------------------------------------------------------|------|
| Distance travelled (km)                                           | 20   |
| Mass of carbon dioxide produced per person travelling by car (kg) | 1.20 |
| Mass of carbon dioxide produced per person travelling by bus (kg) | 0.20 |

**Four** people each travel a total of 20 km per day.  
Use the information to answer the following questions.

- (i) Use the equation below to answer the following questions.

total mass of carbon dioxide produced (kg) = number of people  $\times$  mass of carbon dioxide produced per person

- I. Calculate the total mass of carbon dioxide produced when four people drive 20 km by car. [2]

mass of carbon dioxide produced = ..... kg

- II. Calculate the mass of carbon dioxide produced if four people travelled 20 km by bus. [1]

mass of carbon dioxide produced = ..... kg

- (ii) Calculate the savings in carbon dioxide production per week if they travel by bus on the same 20 km journey from Monday to Friday instead of using their cars. [2]

savings in carbon dioxide production = ..... kg

